In [1]:
```
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
```

In [2]:
```
dataset=pd.read_csv("Social_Network_Ads.csv")
```

In [3]:
```
dataset
```

Out[3]:

	User ID	Gender	Age	EstimatedSalary	Purchased
0	15624510	Male	19	19000	0
1	15810944	Male	35	20000	0
2	15668575	Female	26	43000	0
3	15603246	Female	27	57000	0
4	15804002	Male	19	76000	0
...	...	...	...	...	...
395	15691863	Female	46	41000	1
396	15706071	Male	51	23000	1
397	15654296	Female	50	20000	1
398	15755018	Male	36	33000	0
399	15594041	Female	49	36000	1

400 rows × 5 columns

In [4]:
```
dataset=pd.get_dummies(dataset,drop_first=True)
```

In [5]:
```
dataset
```

Out[5]:

	User ID	Age	EstimatedSalary	Purchased	Gender_Male
0	15624510	19	19000	0	True
1	15810944	35	20000	0	True
2	15668575	26	43000	0	False
3	15603246	27	57000	0	False
4	15804002	19	76000	0	True
...	...	...	...	...	...
395	15691863	46	41000	1	False
396	15706071	51	23000	1	True
397	15654296	50	20000	1	False
398	15755018	36	33000	0	True
399	15594041	49	36000	1	False

400 rows × 5 columns

In [6]:
```
dataset=dataset.drop("User ID",axis=1)
```

In [7]:
```
dataset
```

Out[7]:

	Age	EstimatedSalary	Purchased	Gender_Male
0	19	19000	0	True
1	35	20000	0	True
2	26	43000	0	False
3	27	57000	0	False
4	19	76000	0	True
...	...	...	...	...
395	46	41000	1	False
396	51	23000	1	True
397	50	20000	1	False
398	36	33000	0	True
399	49	36000	1	False

400 rows × 4 columns

In [8]:
```
dataset["EstimatedSalary"].value_counts()
```

Out[8]:

```
EstimatedSalary
72000     12
80000     11
79000     10
71000      9
75000      9
          ..
99000      1
129000     1
46000      1
64000      1
139000     1
Name: count, Length: 117, dtype: int64
```

In [9]:
```
x = dataset[["Age","EstimatedSalary","Gender_Male"]]
y = dataset["Purchased"]
```

In [10]:
```
dataset.shape
```

Out[10]:

```
(400, 4)
```

In [11]:
```
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=1/3,random_state=0)
```

In [13]:
```
from sklearn.ensemble import RandomForestClassifier
clf = RandomForestClassifier(max_depth=2,criterion='gini',max_features='sqrt')
clf.fit(x_train, y_train)
```

Out[13]:

▾ RandomForestClassifier

▸Parameters

▸Fitted attributes

In [14]:
```
y_pred=clf.predict(x_test)
y_pred
```

Out[14]:

```
array([0, 0, 0, 0, 0, 0, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 1,
       0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0,
       1, 0, 0, 1, 0, 1, 1, 0, 0, 1, 1, 1, 0, 0, 1, 0, 0, 1, 0, 1, 0, 1,
       0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 1, 1, 1, 1, 0, 0, 1, 0, 0, 1,
       1, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 1, 1, 0, 0, 1, 1, 0, 0, 0, 0,
       0, 0, 0, 1, 1, 1, 1, 0, 1, 1, 0, 1, 0, 1, 0, 0, 1, 1, 0, 1, 0, 1,
       0, 1])
```

In [15]:
```
from sklearn.metrics import confusion_matrix
cm=confusion_matrix(y_test, y_pred)
print(cm)
```

```
[[79  6]
 [ 8 41]]
```

In [16]:
```
from sklearn.metrics import classification_report
clf_report = classification_report(y_test, y_pred)
print(clf_report)
```

```
              precision    recall  f1-score   support

           0       0.91      0.93      0.92        85
           1       0.87      0.84      0.85        49

    accuracy                           0.90       134
   macro avg       0.89      0.88      0.89       134
weighted avg       0.89      0.90      0.90       134
```

In [ ]: